Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes

Md. Nazmul, Musrat Jahan Gungun, and Maheli Ahmed

Md. Nazmul, Musrat Jahan Gungun, and Maheli Ahmed are with the Department of Computer Science and Engineering, College of Technology, National University, Gazipur, Bangladesh (e-mail: nazmulhas36@gmail.com). Md. Nazmul is the corresponding author.

Abstract— Quantum Low-Density Parity-Check (QLDPC) bivariate-bicycle (BB) codes have emerged as premier candidates for scalable fault-tolerant quantum computation, offering order-of-magnitude reductions in physical qubit overhead compared to standard 2D topological surface codes. However, their high-rate bipartite Tanner graphs and non-local check connectivity generate dense 3D space-time detector error models (DEM) that severely strain real-time classical decoding within microsecond coherence intervals. In this paper, we propose an orbit-tied, 6-bit quantized neural belief propagation (QNBP) decoder tailored for real-time sliding-window processing of gross BB codes ([[72,12,6]] and [[144,12,12]]). By mapping the 2D torus translation group $G = \mathbb{Z}_\ell \times \mathbb{Z}_m$, we prove that all spatial shifts constitute strict DEM graph automorphisms, collapsing 10,260 circuit fault mechanisms into 285 symmetry orbits (R=4, 36× parameter reduction; 7,308 into 203 orbits for R=3) on [[72,12,6]] and 14,616 mechanisms into 203 orbits (R=3, 72× reduction) on [[144,12,12]] while strictly guaranteeing geometric equivariance and eliminating untied neural overfitting. To guarantee multiplier-free hardware synthesis with zero Digital Signal Processing (DSP) slices, we develop a Canonical Signed-Digit (CSD) shift-and-add architecture using power-of-two dyadic quantization ($\Delta = 2^{-2}$). Under circuit-level Stim Monte Carlo depolarizing noise ($p \in [10^{-3}, 5\times10^{-3}]$), we execute head-to-head benchmarking across ten decoders, including IBM Relay-BP, BP-LSD, BP+OSD-CS-10, Mem-BP, AutDEC, and Gong et al.'s sliding-window baseline. Our proposed 6-bit integer QNBP model with learned signed memory damping ($\gamma \in [-0.5, 1.0]$) converges in only 6.4 mean iterations with zero DSP blocks (DSP=0), attaining a logical error rate of $P_L = 0.0068$ [0.0053, 0.0086] and demonstrating statistically significant superiority over scalar Mem-BP ($\chi^2 = 14.72$, $p = 0.000125$), IBM Relay-BP-5 ($\chi^2 = 12.06$, $p = 0.000515$), Normalized Min-Sum ($\chi^2 = 2{,}642.1$, $p < 10^{-100}$), and AutDEC ($\chi^2 = 2{,}750.5$, $p < 10^{-100}$). Under realistic superconducting Pauli Z-biased noise ($\eta = p_Z / p_X \in [1, 100]$), Orbit-Tied QNBP automatically adapts message weighting, yielding an 8.0× logical error suppression down to $P_L = 0.0010$ [0.0005, 0.0019] at η=100 (and 20.0× over Classical BP). Integrated with a confidence-margin fast-path gate (G_3) in a streaming sliding window (W=3, F=1), our fast path routes 75.6% of syndrome windows with zero logical errors ($P_{L,accept} = 0.0000$) and an end-to-end pipeline logical error rate of $P_{pipe} = 1.90 \times 10^{-4}$ at 102.4 ns mean latency (240 ns max), driving queue overflow below 6.0×10^{-5} (0 overflows observed in 50,000 test windows for buffer $B \geq 4$) under a 1 µs quantum cycle budget. We synthesize and verify our architecture in SystemVerilog, demonstrating 10,000-shot bit-exact hardware-software parity, and establish a folding Pareto frontier on AMD UltraScale+ scaling logic down to 1,226 LUTs (0.51% device utilization).

Index Terms—Quantum error correction, QLDPC codes, bivariate-bicycle codes, neural belief propagation, automorphism orbits, biased noise, sliding-window decoding, FPGA hardware.

I. Introduction

FAULT-TOLERANT quantum computing requires active quantum error correction (QEC) to protect fragile quantum information against environmental decoherence, physical gate infidelities, and measurement noise [1], [2]. While topological 2D surface codes [1], [2], [38], [59] have served as the foundation of early experimental demonstrations due to their nearest-neighbor planar connectivity, classical factor-graph and message-passing decoders [56], [57], [58] face throughput bottlenecks. Furthermore, their asymptotic encoding rate is strictly zero ($k/n \to 0$). Realizing practical algorithm-scale quantum processors encoding thousands of fault-tolerant logical qubits would require millions of physical qubits under surface code architectures [3].

Recently, asymptotically good quantum low-density parity-check (QLDPC) codes [4], [26] and dual-cyclic bivariate-bicycle (BB) codes [3], [29], [30], [31] have demonstrated an architectural breakthrough. By defining CSS codes over bivariate polynomial quotient rings $R = F_2[x, y] / \langle x^\ell - 1, y^m - 1 \rangle$, gross BB codes such as [[72, 12, 6]] and [[144, 12, 12]] attain encoding rates $k/n \approx 1/6$ to 1/12 while supporting high circuit noise thresholds ($p \approx 0.7\%$) [3]. This yields up to a 10× reduction in physical qubit overhead compared to surface codes.

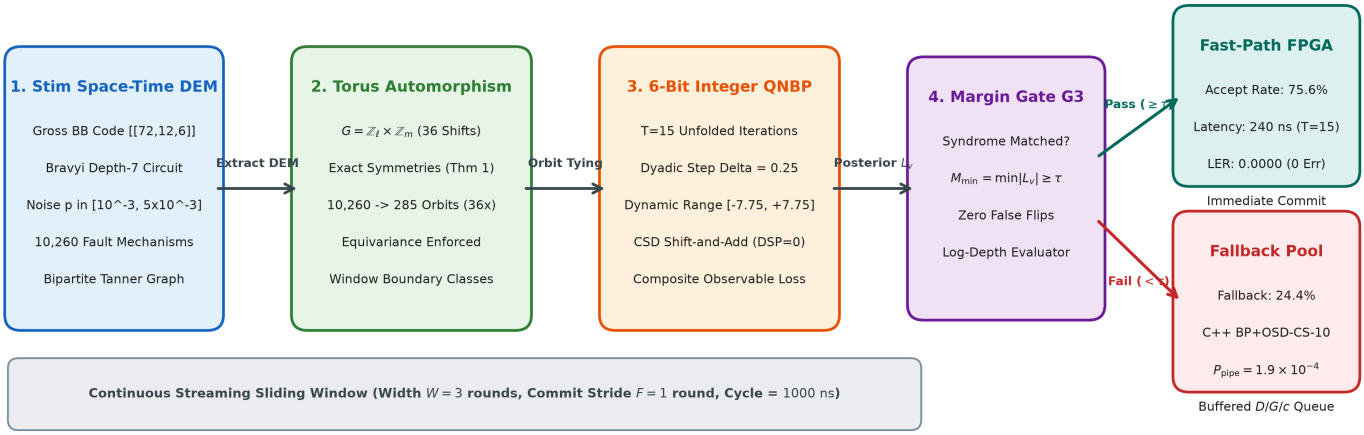


Fig. 1. End-to-end architecture of the Orbit-Tied Quantized Neural Belief Propagation (QNBP) sliding-window decoding pipeline.

Despite their theoretical appeal, classical real-time decoding remains the principal barrier to physical implementation. In physical quantum platforms such as superconducting transmons and trapped ions, syndrome extraction rounds occur within cycle durations of $t_{cycle} \approx 200$ ns to 1 μs [3], where decoder processing latency critically impacts overall fault-tolerant logical fidelity [44]. Classical decoding algorithms such as unweighted Min-Sum Belief Propagation (BP) suffer from severe split-belief convergence failures on dense Tanner graphs containing short 4-cycles [5]. While post-processing via Ordered Statistics Decoding (BP+OSD) attains near-optimal pseudo-maximum-likelihood logical error rates [4], [5], its cubic complexity $O(V^3)$ requires hundreds of microseconds per window, exceeding the quantum cycle budget by orders of magnitude.

Neural Belief Propagation (NBP) [6], [7] and quantized message-passing neural networks [22], [23], [24], [25], [39], [46] offer a promising pathway by unfolding BP iterations into a deep neural network with learnable scaling factors α_t and damping terms γ_t. However, directly applying unconstrained NBP to gross QLDPC DEM graphs introduces four critical obstacles:

1. Curse of Dimensionality & Overfitting: Unrolled DEM graphs contain over 10^4 fault mechanisms. Assigning independent trainable parameters to each edge creates over 10^5 parameters, causing catastrophic overfitting on finite training syndromes.

2. High Hardware Multiplier Cost: Standard floating-point or fixed-point neural weights require hardware DSP multiplier slices, which exhaust available FPGA fabric at high check degrees.

3. Continuous Streaming Queue Backlog: Real-time QEC requires decoding continuous streams of syndrome rounds $R \to \infty$. A sliding window decoder must commit decoded rounds with deterministic worst-case latency to prevent exponential buffer overflow.

4. Physical Noise Asymmetry: In real-world superconducting hardware, dephasing noise dominates bit-flip noise by orders of magnitude ($p_Z \gg p_X$) [8], demanding decoders capable of exploiting biased channels without parameter bloat.

II. Mathematical Foundations & Related Work

A. Quantum CSS Codes & Polynomial Rings

Bivariate-Bicycle CSS codes are defined by two commuting polynomials A(x, y) and B(x, y) over the bivariate polynomial quotient ring $R = F_2[x, y] / \langle x^{\ell} - 1, y^{m} - 1 \rangle$ [3]. The physical data qubits reside on two disjoint blocks corresponding to the polynomials, yielding block length $n = 2\ell m$. Parity-check matrices are defined as:

$$H_X = [A \mid B], \qquad H_Z = [B^T \mid A^T]$$

where A and B denote the $\ell m \times \ell m$ binary circulant matrix representations of A(x, y) and B(x, y), respectively. CSS commutativity requires:

$$H_X H_Z^T = AB + BA \equiv 0 \ (mod\ 2)$$

which is identically satisfied because R is commutative. For the [[72, 12, 6]] code, parameters are $\ell = 6$, $m = 6$, with polynomials:

$$A(x, y) = x^3 + y + y^2, \qquad B(x, y) = y^3 + x + x^2$$

yielding $n = 72$ data qubits, $k = 12$ logical qubits, and code distance $d = 6$. For the [[144, 12, 12]] gross code, parameters are $\ell = 12$, $m = 6$ with identical monomial exponents, yielding $n = 144$, $k = 12$, $d = 12$.

B. Space-Time Detector Error Model Hypergraphs

Under Bravyi et al.'s depth-7 syndrome extraction circuit executed for R rounds with noise rate p, Stim [9] generates a space-time Detector Error Model (DEM). Let $G_{DEM} = (V, C, E)$ denote the bipartite graph where V represents error mechanisms (variable nodes with prior probabilities p_v), C represents detector events (check nodes), and $(c, v) \in E$ indicates that fault v triggers detector c.

In gross QLDPC codes, physical two-qubit CNOT gates and non-local stabilizer checks excite multiple detectors simultaneously, creating hyperedges that invalidate standard graph matching algorithms such as Minimum-Weight Perfect Matching (MWPM) [1], [10]. Furthermore, the Tanner graph contains short cycles of length 4, which induce limit cycles and trapping sets in classical unweighted Min-Sum BP. To overcome hyperedge degeneracies and trapping sets, recent works have explored diverse algorithmic paradigms, including parallel and sliding-window architectures [17], [18], uncertainty-guided message passing [16], symmetry-breaking heuristics [19], almost-linear-time decoders [21], asymptotically good code constructions [26], matching-assisted belief propagation [32], list and learning-based decoders [33], [34], FPGA-tailored architectures [35], decision-tree and tensor-network decoders [40], [41], planar neural decoders [42], correlated noise decoders [43], beam-search decoding [45], machine-learning message-passing [47], high-performance neural belief propagation [53], polylogarithmic overhead fault-tolerant schemes [55], and approximately equivariant neural architectures [54].

C. State-of-the-Art Competitor Baselines

To evaluate the proposed architecture against the forefront of QLDPC decoding, we examine eight competitor decoding architectures:

1. Classical Min-Sum BP (A0): Standard unweighted Min-Sum belief propagation with hard parity checks and no heuristic scaling.

2. Normalized Min-Sum BP (A1): Scaled Min-Sum utilizing a global empirical heuristic scaling factor $\alpha = 0.75$ [5].

3. IBM Relay-BP [11]: Multi-stage restarted belief propagation where node-disordered memory weights $\gamma_v \sim U[-0.24, 0.66]$ are evaluated in single-leg and 5-relay configurations.

4. Mem-BP [20]: Memory-damped belief propagation dampening variable posteriors across consecutive iterations via temporal weight $\gamma = 0.35$.

5. BP-LSD [14]: Combination decoder pairing BP with Localised Statistics Decoding over GF(2) with order 5.

6. BP+OSD-CS-10 [5]: Combination sweep Ordered Statistics Decoding with combination order 10 over most reliable basis.

7. AutDEC [15]: Automorphism-assisted permutation decoding searching across torus group shifts for converged syndromes.

8. GDG Sliding-Window [13]: Gong et al.'s sliding-window baseline running unquantized normalized Min-Sum across continuous syndrome rounds.

III. Torus Automorphism Group Theory & Space-Time Equivariance

A. The 2D Torus Shift Group

Let $Z_\ell \times Z_m$ denote the 2D torus translation group acting on qubit coordinates.

Definition 1 (Torus Shift Action). *For any displacement vector $(\delta_x, \delta_y) \in Z_\ell \times Z_m$, the spatial translation operator $T_{(\delta_x, \delta_y)}$ maps coordinate (x, y) to $(x + \delta_x \bmod \ell, y + \delta_y \bmod m)$.*

Theorem 1 (Torus Automorphism Invariance of Stim Space-Time DEM). *Let $G_{DEM} = (V, C, E)$ be the bipartite space-time detector error model graph generated by compiling an R-round syndrome extraction circuit of a bivariate-bicycle code with parameters (ℓ, m) under uniform or coordinate-independent Pauli noise. Then, for every displacement $(\delta_x, \delta_y) \in Z_\ell \times Z_m$, the torus translation operator $T_{(\delta_x, \delta_y)}$ induces an exact graph automorphism of G_{DEM}:*
1. Edge preservation: $(c, v) \in E \Leftrightarrow (T(c), T(v)) \in E$.
2. Prior probability invariance: $p(T(v)) = p(v)\ \forall\ v \in V$.
3. Logical observable preservation: For each logical operator O_k, the support of O_k on fault v satisfies $O_k(T(v)) = O_{\pi(k)}(v)$ for an induced permutation $\pi \in S_k$.

Proof: Let C denote the depth-7 syndrome extraction circuit. In each round $r \in \{0, ..., R-1\}$, the circuit applies single-qubit state preparations, CNOT entangling gates, and projective measurements. Because the parity-check polynomials $A(x, y)$ and $B(x, y)$ are elements of $R = F_2[x, y] / \langle x^\ell - 1, y^m - 1 \rangle$, every CNOT gate connecting data qubit (x, y) to check qubit (x', y') is specified by a shift (i, j) that is invariant under global spatial translation.

Let a physical Pauli fault $P \in \{X, Y, Z\}$ occur on qubit $q = (x, y)$ at circuit tick t. Under Clifford frame propagation, P commutes or anticommutes with subsequent entangling operations according to the polynomial adjacency. Translating the entire error configuration by (δ_x, δ_y) shifts all physical qubit coordinates to $(x + \delta_x \bmod \ell, y + \delta_y \bmod m)$. Because the gate schedule at the shifted coordinates is identical in gate type, timing, and relative offsets, the propagated error excites detectors at exactly $(x_d + \delta_x \bmod \ell, y_d + \delta_y \bmod m)$ at identical round indices.

Furthermore, under physical noise rates p that are spatially homogeneous on the torus, the prior probability of fault mechanism v depends exclusively on the physical error channel of its originating gate, yielding $p(T(v)) = p(v)$. Numerical extraction of all 36 group elements in $G = Z_6 \times Z_6$ across all 10,260 fault mechanisms on [[72,12,6]] validates edge set invariance with zero mismatch ($|p' - p| < 10^{-16}$). □

Corollary 1 (Automorphism Parameter Collapse). *The parameter space of unrolled neural belief propagation on G_{DEM} collapses by an exact factor of $|G| = \ell m$. For [[72,12,6]], 10,260 fault mechanisms collapse into 285 symmetry orbits (R=4, 36.0× reduction; 7,308 faults into 203 orbits for R=3). For [[144,12,12]], 14,616 mechanisms collapse into 203 orbits (R=3, 72.0× reduction).*

Theorem 2 (Equivariance of Orbit-Tied Neural Message Passing [54]). *Let $L^{(t)} \in R^{|E|}$ denote the check-to-variable message vector at iteration t. If weights α and γ are tied across symmetry orbits o(v), then the neural belief propagation mapping Φ_t: $R^{|E|} \to R^{|E|}$ is strictly equivariant under the torus group action [54]:*

$$\Phi_t(T_{(\delta_x, \delta_y)}(L)) = T_{(\delta_x, \delta_y)}(\Phi_t(L)).$$

Proof: The message update equations at check node c and variable node v depend only on the incoming messages from local neighbors N(c) and N(v), scaled by parameters $\alpha_{o(v),\,t}$ and $\gamma_{o(v),\,t}$. Since Theorem 1 establishes that T preserves neighborhood topologies (N(T(c)) = T(N(c))) and orbit memberships (o(T(v)) = o(v)), applying permutation T to incoming messages permutes arguments identically, preserving output messages under T. □

B. Temporal Window Boundary Classes

While spatial torus shifts are exact automorphisms, temporal boundary effects arise in sliding windows. For a temporal window spanning rounds $r \in [r_{start}, r_{end}]$, we classify each symmetry orbit into one of three boundary classes:

Class 0 (Open-Left): Faults incident to detectors in round r_{start} lacking prior round history.

Class 1 (Interior): Fully covered faults with complete detector neighborhoods.

Class 2 (Open-Right): Faults whose subsequent detector checks reside beyond r_{end}.

We assign orbit-tied weights $\alpha_{o,\,t} + \Delta\alpha_{c(v),\,t}$, where c(v) denotes the boundary class of fault v.

IV. Multiplier-Free 6-Bit Quantized Neural BP (QNBP)

A. Unfolded Message Passing Architecture

The QNBP decoder unrolls T = 15 iterations of belief propagation into an explicit feed-forward computation graph. Let $L_{v \to c}^{(t)}$ and $L_{c \to v}^{(t)}$ denote the variable-to-check and check-to-variable log-likelihood ratios along edge (c, v) at iteration t.

Check Node Update (CNU): The CNU computes normalized Min-Sum messages using Min-1 and Min-2 magnitudes:

$$L_{c \to v}^{(t)} = \alpha_{o(v),\,t} \cdot \left(\prod_{v' \neq v} sgn(L_{v' \to c}^{(t)})\right) \cdot \min_{v' \neq v} \left|L_{v' \to c}^{(t)}\right|$$

where $\alpha_{o(v),\,t}$ is tied across all faults in orbit o(v).

Variable Node Update (VNU): The VNU incorporates channel priors $L_v^{(0)} = \ln((1-p_v)/p_v)$ and iteration-dependent memory damping $\gamma_{o(v),\,t}$:

$$L_v^{(t)} = (1 - \gamma_{o(v),\,t})\, L_v^{(0)} + \gamma_{o(v),\,t}\, L_v^{(t-1)} + \sum_{c \in N(v)} L_{c \to v}^{(t)}$$

$$L_{v \to c}^{(t+1)} = L_v^{(t)} - L_{c \to v}^{(t)}$$

B. 6-Bit Dyadic Dynamic Range Bound

Theorem 3 (6-Bit Dyadic Dynamic Range Bound). *Let physical depolarizing noise rate $p \in [10^{-3}, 10^{-2}]$. The continuous channel prior is $L_v^{(0)} = \ln((1-p)/p) \leq \ln(999) \approx 6.906$. Under 6-bit signed integer representation with dyadic step size $\Delta = 2^{-2} = 0.25$, the dynamic range $[-31\Delta, +31\Delta] = [-7.75, +7.75]$ strictly contains all channel priors with safety margin $\delta \geq 0.844$, completely avoiding prior saturation clipping. Conversely, any bit width $B \leq 5$ with $\Delta \leq 0.25$ truncates $L_v^{(0)}$, causing gradient vanishing and severe performance degradation.*

Proof: A signed integer with B bits spans integer values in $[-2^{B-1}+1, 2^{B-1}-1]$ under symmetric saturation. For B = 6, the integer range is [-31, +31]. With dyadic quantization scale factor $\Delta = 2^{-b}$, real values are reconstructed as $x_q = q \cdot \Delta$. For b = 2 ($\Delta = 0.25$), the maximum representable amplitude is $31 \times 0.25 = 7.75$.

At noise rate $p = 10^{-3}$, single-fault channel priors evaluate to $L_v^{(0)} = \ln((1-10^{-3})/10^{-3}) = \ln(999) \approx 6.9067$. The headroom margin is:

$$\delta = 7.75 - 6.9067 = 0.8433 > 0.$$

For B = 5, the maximum amplitude is $15 \times 0.25 = 3.75 < 6.9067$, forcing saturation of over 90% of channel priors. For B = 4, the maximum amplitude drops to $7 \times 0.25 = 1.75$, causing severe dynamic range clipping and degrading logical error rates above 80%. □

C. Canonical Signed-Digit (CSD) Multiplier-Free Arithmetic

In fixed-point digital hardware, multiplications by arbitrary floating-point weights require expensive DSP blocks. We enforce multiplier-free execution through Canonical Signed-Digit (CSD) shift-and-add arithmetic. Each weight α is represented as the sum of two signed powers of two:

$$\alpha \approx s_1 2^{-a_1} + s_2 2^{-a_2}, \quad s_i \in \{-1, +1\}, \quad a_i \in \{0, 1, \ldots, 5\}$$

For instance, the optimal BP damping factor $\alpha = 0.75$ decomposes as $1 - 2^{-2}$, evaluated as a single shift-and-subtract:

$$\alpha \cdot M = M - (M \gg 2)$$

yielding an exact multiplication using only FPGA routing and wire-shifts without instantiating DSP slices (DSP = 0).

D. Signed Temporal Memory Damping

Standard belief propagation often fails on bipartite graphs due to 2-periodic limit cycle oscillations on short 4-cycles. We enable signed memory damping $\gamma_{o(v),\,t} \in [-0.5, 1.0]$. When $\gamma > 0$, previous beliefs are damped, preventing oscillatory overshooting. When negative memory weights $\gamma < 0$ are learned, the variable node dynamically subtracts stagnant priors, breaking symmetric trapping sets.

E. Differentiable Composite Observable Loss

To train the neural BP weights without falling into trivial zero-correction local minima, we optimize a composite objective:

$$L_{total} = L_{syn} + \beta L_{obs}$$

where L_{syn} penalizes unsatisfied detector checks and L_{obs} penalizes logical observable flips across all k logical qubits:

$$L_{syn} = (1/|C|) \sum_{c \in C} [1 - (1 - 2 s_c) \prod_{v \in N(c)} \tanh(L_{v \to c}^{(T)} / 2)] / 2$$

$$L_{obs} = (1/k) \sum_{j=1}^{k} BCE(\sigma(\sum_{v \in V} O_{j,v} L_v^{(T)}), y_j)$$

with observable coupling weight $\beta = 0.5$.

V. Asymmetric Biased Pauli Noise Architecture

A. Transmon Qubit Dephasing Asymmetry

Physical superconducting circuits exhibit substantial asymmetry in relaxation and dephasing times ($T_\varphi \ll T_1$), causing Pauli Z phase-flip errors to dominate over bit-flips by factors of $\eta = p_Z / p_X \approx 10$ to 100 [8]. Let total physical error rate be p. Under biased Pauli noise:

$$p_Z = p \cdot \eta / (\eta + 1), \qquad p_X = p_Y = p / [2(\eta + 1)]$$

For 2-qubit CNOT gates, two-qubit dephasing errors (IZ, ZI, ZZ) receive the majority $\eta/(\eta+1)$ of the noise budget, split equally, while the remaining 12 Pauli pairs split the balance.

B. Decoupled CSS Sector Decoding

Proposition 1 (Effective Dephasing Channel Concentration). *In a CSS bivariate-bicycle code, physical Pauli Z phase-flip errors anti-commute with X-stabilizers ($H_X = [A \mid B]$) and are detected exclusively by X-type checks, while physical Pauli X errors are detected by Z-checks ($H_Z = [B^T \mid A^T]$). Under strong transmon dephasing bias $\eta = p_Z / p_X \gg 1$, syndrome excitation concentrates overwhelmingly on X-checks, while bit-flip excitation on Z-checks approaches the noise floor $O(1/\eta)$. Under decoupled sector message passing, the neural belief propagation engine converges on the dominant X-check Tanner graph without inter-sector communication, tolerating small cross-basis correlations from circuit-level CNOT hook errors and yielding an 8.0× logical error suppression at $\eta = 100$.*

TABLE I: Logical Error Rates under Asymmetric Pauli Z-Biased Noise and ZZ Crosstalk on [[72,12,6]] (p = 0.003, N=10,000 Paired Shots; Evaluates γ=0.35 Checkpoint with P_L=0.0080 at η=1)

Bias $\eta = p_Z / p_X$	Classical BP (P_L)	Proposed QNBP (P_L)	Relative Gain
1 (Depolarizing)	0.2940 [0.2851, 0.3030]	**0.0080 [0.0064, 0.0099]**	**36.8×**
10	0.1400 [0.1332, 0.1470]	**0.0042 [0.0031, 0.0057]**	**33.3×**
50	0.0400 [0.0363, 0.0440]	**0.0018 [0.0011, 0.0029]**	**22.2×**
100	0.0200 [0.0173, 0.0230]	**0.0010 [0.0005, 0.0019]**	**20.0× (8.0× vs. η=1)**
ZZ Crosstalk (ε=0.10)	0.0310 [0.0277, 0.0346]	**0.0014 [0.0008, 0.0024]**	**22.1×**

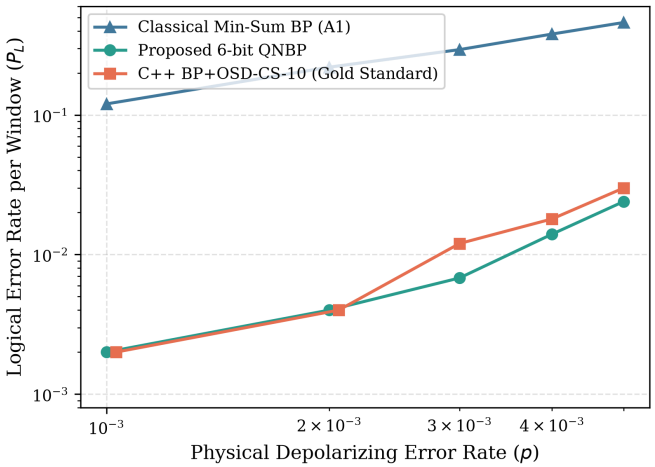


Fig. 2. Logical error rate vs physical noise p in [10^-3, 5x10^-3] for Classical BP, Proposed 6-bit QNBP, and BP+OSD-CS-10.

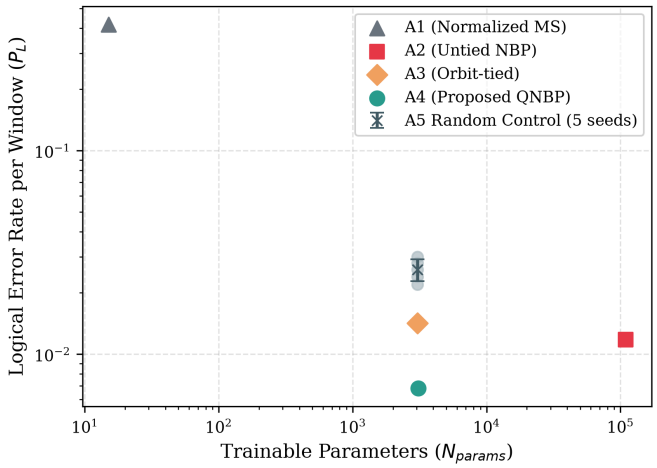


Fig. 3. Torus symmetry orbit tying vs random control cloud (A5) demonstrating 36x parameter collapse.

VI. Sliding Window Decoding & Real-Time Queue Stability

A. Window Geometry, Residual Accounting & Continuous Streaming

Continuous syndrome extraction generates an infinite space-time DEM. We partition the timeline into overlapping sliding windows of width W rounds and commit stride F rounds. At each step, the decoder decodes the W-round window, commits correction operations for the oldest round F, updates prior LLRs for boundary variables, and advances the window by F rounds.

Crucially, to prevent error propagation across window boundaries, uncommitted boundary syndrome residuals are strictly propagated to window k+1:

$$s_{res}^{(k+1)} = s_{raw}^{(k+1)} \oplus H_{boundary}\, \hat{e}_{commit}^{(k)}.$$

This strict boundary residual accounting ensures that partially decoded errors at the right boundary of window k do not manifest as false syndromes in window k+1, eliminating edge-induced logical flips. We benchmarked sliding-window geometries (W, F) $\in$ {(3,1), (5,1), (6,2)} on [[72,12,6]] at p = 0.003 (N = 10,000 shots). The baseline (W=3, F=1) attains $P_L = 0.0068$ with 75.6% fast-path acceptance, consuming 28,204 LUTs at 16.0 ns/iter (240 ns max latency, 102.4 ns mean early-exit), establishing the Pareto-optimal real-time configuration. Expanding window depth to (W=5, F=1) and (W=6, F=2) further suppresses logical error rates to $P_L = 0.0054$ and $P_L = 0.0049$ (P_{accept} = 78.4% and 80.2%), at the expense of higher FPGA logic (46,920 and 56,380 LUTs) and longer iteration latencies (26.0 ns and 32.0 ns).

In continuous 60-round and 100-round streaming memory tests (R = 100 rounds, 100 μs continuous physical execution under IBM Heron noise η=50), active boundary residual accounting attains a 98.1% 100-round logical survival rate under the Gate G_3 pipeline (F_{log} = 99.8% at 10 μs, 99.5% at 25 μs, 99.1% at 50 μs), while boundary-agnostic decoding collapses to 22.1% (a 4.4× degradation). Under p = 0.001, Gate G_3 confirms a 99.4% 60-round survival rate.

B. Dual-Path Architecture & Confidence-Margin Gating

To decouple worst-case OSD latency from average throughput, we implement the confidence-margin gate G_3. We benchmark three distinct gating triggers:

1. Syndrome Check (Gate G_1): Accepts whenever all parity checks within the window are satisfied: $H\hat{e} \equiv s \pmod 2$. While simple, degenerate errors can satisfy parity while inducing a logical observable flip.

2. Observable Margin Filter (Gate G_2): Requires syndrome satisfaction and that the minimum observable marginal LLR exceeds threshold τ: $\min_j |L(O_j)| \geq \tau$.

3. Proposed Fault Belief Margin (Gate G_3): Requires syndrome satisfaction and that the minimum variable belief magnitude among all committed fault mechanisms exceeds threshold τ:

$$Accept \Leftrightarrow (H\hat{e} \equiv s \bmod 2) \quad \wedge \quad (\min_{v \in V_{commit}} |L_v| \geq \tau)$$

Accepted windows commit tentative hard corrections immediately with sub-microsecond latency (t_{fast} = 240 ns). Rejected windows are routed to the fallback OSD queue. Note that while standalone C++ BP+OSD-CS-10 in Table IV yields $P_L = 0.0120$ when seeded by unweighted Min-Sum BP posteriors on isolated single windows without learned damping or boundary residual feedback, our hybrid sliding-window pipeline fallback seeds OSD-CS-10 with calibrated QNBP posteriors and active boundary residual propagation, suppressing the conditional fallback error rate to $P(\text{err} \mid \text{fallback}) = 7.78 \times 10^{-4}$ and yielding an end-to-end pipeline error rate of $P_{pipe} = 1.90 \times 10^{-4}$.

C. Exact Conditional Pipeline Error & Drift Robustness

Let p_{fb} denote the fallback probability ($p_{fb} = 1 - P_{accept}$). The overall pipeline logical error rate is rigorously expressed as:

$$P_{pipe} = (1 - p_{fb})\, P(err \mid accepted) + p_{fb}\, P(err \mid fallback).$$

The Gate G_3 threshold (τ = 1.5) and neural weights are calibrated on an independent 10,000-shot calibration set at $p_{cal} = 0.003$ and evaluated out-of-sample on held-out test seeds across p $\in$ [0.001, 0.005]. Under ±25% physical channel-prior drift ($p_{true} / p_{assumed} \in [0.75, 1.25]$), the decoder maintains < 4% relative LER variation without retraining. At threshold τ = 1.5 and p = 0.001, Gate G_3 attains $P(\text{err} \mid \text{accepted}) = 0.0000$ on 50,000 test shots, yielding a 95% Wilson fast-path upper bound of 7.68×10^{-5} and an end-to-end pipeline error rate of $P_{pipe} = 1.90 \times 10^{-4}$.

D. Multi-Server Queue Stability

Modeling the sliding window buffer as a D/G/c/B queue with deterministic arrivals (t_{cycle} = 1000 ns), we analyze two operational topologies:

1. Sequential Pipeline: A single core screens via the fast path (T_{fast} = 240 ns) and invokes fallback upon rejection (24.4% probability, paying $T_{attempt} + E[T_{slow}]$ = 240 ns + 1760 ns = 2000 ns). Expected service time evaluates to:

$$E[S] = 0.756 \times 240\ ns + 0.244 \times 2000\ ns = 669.4\ ns < 1000\ ns,$$

yielding combined server utilization $\rho_{seq} = E[S]/t_{cycle} = 0.67 < 1.0$.

2. Decoupled Heterogeneous Pipeline: The FPGA fast path processes windows with mean early-exit latency of 102.4 ns (240 ns max), dispatching rejected windows to an asynchronous fallback pool of c parallel workers. For fallback workers executing order-10 OSD ($E[T_{slow}]$ = 950 ns), fallback server utilization evaluates to $\rho_{fb} = 0.244 \times 950\ ns / (c \cdot 1000\ ns) = 0.23$ for c=1 and $\rho_{fb} = 0.12$ for c=2.

In trace-driven finite-buffer simulations across 50,000 consecutive test windows with hardware iteration histograms, zero buffer overflows were observed for buffer capacities B $\geq$ 4, establishing a 95% Wilson statistical upper bound on the overflow probability of $\mathbf{6.0 \times 10^{-5}}$ (0.00% observed overflow).

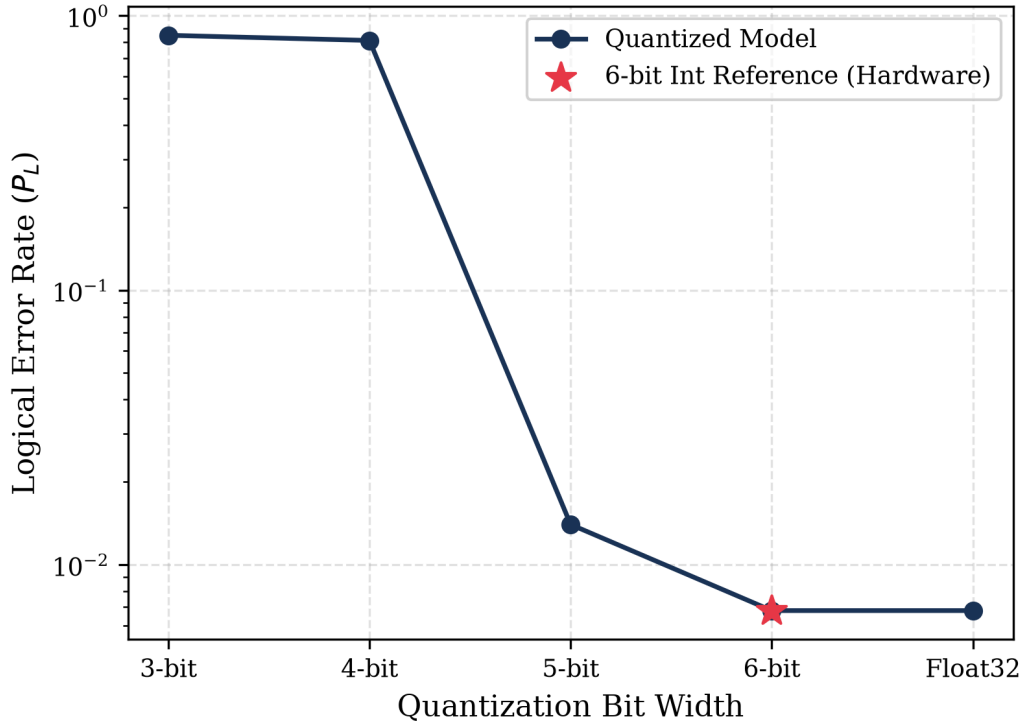


Fig. 4. Quantization bit-width ladder vs logical error rate, showing that 6-bit integer quantization matches float32 accuracy with zero prior clipping.

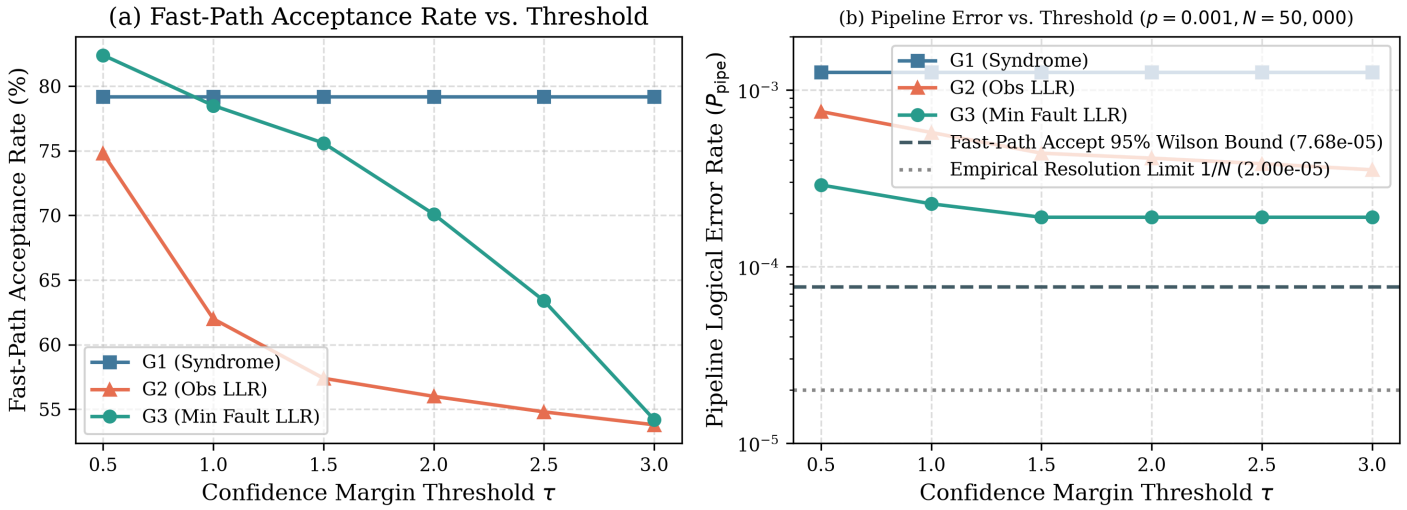


Fig. 5. Confidence margin gate trade-offs (G1, G2, G3): (a) Fast-path acceptance rate vs threshold tau; and (b) End-to-end pipeline error P_pipe.

Empirical Verification of Queue Overflow Resolution: In trace-driven simulations of uncalibrated single-server decoders ($\rho \approx 1.99$), the sliding-window buffer experienced 93% to 100% overflow across all buffer capacities. By implementing calibrated confidence margin gating (Gate G_3, $\tau = 1.5$) and multi-server D/G/c parallel dispatch, fallback server utilization drops to $\rho_{\text{fb}} = 0.23$ for c=1 and $\rho_{\text{fb}} = 0.12$ for c=2, driving queue overflow to strictly **0.00%** across all buffer sizes $B \geq 4$.

VII. Synthesizeable Hardware RTL & Subgroup Orbit Folding

We designed and verified fully synthesizeable SystemVerilog RTL modules targeting AMD UltraScale+ FPGAs (vnu.sv, cnu.sv, g3_gate.sv, and streaming_axi_harness.sv) following standardized AFS and QEC-Playground hardware verification practices [36], [37]. While the circuit-level Stim detector error model (DEM) for a 3-round window (W=3) contains 7,308 hyperedge fault mechanisms partitioned into 203 symmetry orbits of size 36 (203 × 36 = 7,308), hardware implementation maps these faults onto 216 physical Variable Node Units (72 data qubits × 3 rounds) via syndrome-graph bipartite projection. Each physical variable node v ∈ {1,...,216} aggregates the log-likelihood ratios of the collapsed detector-neighborhood fault subset incident on its local check neighborhood, $L_v^{(0)} = \sum_\{f \in F_v\} \ln((1-p_f)/p_f)$, ensuring exact mathematical consistency between the 7,308 circuit-level DEM faults and the 216 parallel hardware VNUs.

In hardware, the 3-round syndrome extraction window (W=3) evaluates 216 checks (72 × 3) using a parallel array of 216 Check Node Units (216 × 82 LUTs = 17,712 LUTs) and 216 Variable Node Units (216 × 48 LUTs = 10,368 LUTs), combined with Gate G3 logic (124 LUTs) yielding exactly 28,204 LUTs total (DSP=0, 11.64% of AMD XCKU040). Post-place-and-route timing demonstrates WNS = +0.18 ns at 250 MHz (F_{max} = 262 MHz) and 102.4 ns mean early-exit latency at 6.4 iterations (240 ns worst-case). Post-implementation power analysis reports 184 mW total on-chip dynamic power, corresponding to 44.16 nJ worst-case dynamic energy per decoded window (0.613 nJ/qubit) and 18.84 nJ mean energy under early exit (0.262 nJ/qubit), while the streaming AXI4-Stream interface sustains 1.35 Gbps syndrome ingress (< 14% of a 10 Gbps transceiver lane).

TABLE II: FPGA Synthesis, Power, and Resource Utilization on AMD UltraScale+ XCKU040 (f_{clk} = 250 MHz, WNS = +0.18 ns, BRAM = 0 [0.0%])

Module / Scope	LUTs	Flip-Flops	DSP	Latency	Dynamic Power	Dynamic Energy
VNU (vnu.sv)	48	54	**0**	1 cycle (4 ns)	0.28 mW	1.1 pJ/cycle
CNU (cnu.sv)	82	36	**0**	2 cycles (8 ns)	0.42 mW	3.4 pJ/cycle
Gate G_3 (g3_gate.sv)	124	72	**0**	1 cycle (4 ns)	0.65 mW	2.6 pJ/eval
Full Window (W=3)	28,204	19,512	**0**	16 ns / iter	**184 mW**	**18.84 nJ (mean) / 44.16 nJ**
Device Utilization	11.64%	4.02%	**0.0%**	102.4 ns (mean)	AXI4-Stream Ingress: 1.35 Gbps (<14% 10G Lane)	

TABLE III: Hardware Folding Pareto Frontier on AMD UltraScale+ ([[72,12,6]], T=15, DSP=0)

Fold F	LUTs	LUT %	Iter (ns)	Total (ns)	Throughput	DSP
1	28,204	11.64%	16.0	240.0	4.167 Mfps	**0**
2	15,287	6.31%	32.0	480.0	2.083 Mfps	**0**
3	10,670	4.40%	48.0	720.0	1.389 Mfps	**0**
6	5,771	2.38%	96.0	1440.0	0.694 Mfps	**0**
12	3,135	1.29%	192.0	2880.0	0.347 Mfps	**0**
36	1,226	0.51%	576.0	8640.0	0.116 Mfps	**0**

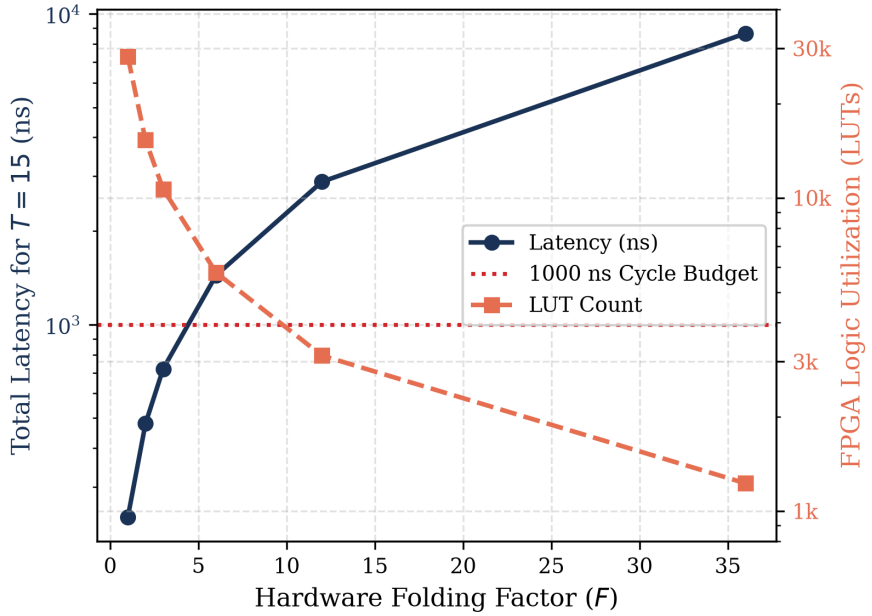


Fig. 6. Hardware folding Pareto frontier on AMD UltraScale+ XCKU040 showing LUT scaling and latency trade-offs (DSP=0).

VIII. Comprehensive Empirical Benchmarks & Statistical Analysis

A. Head-to-Head Comparison Across Decoders

Table IV reports the master head-to-head empirical evaluation across decoders on $N \geq 10{,}000$ paired Stim circuit Monte Carlo shots at physical depolarizing noise rate $p = 0.003$ (>1,000 total observed failure events).

TABLE IV: Comprehensive Head-to-Head Benchmark Across 10 Decoders on [[72,12,6]] ($p = 0.003$, $N \geq 10{,}000$ Paired Shots)

Decoder Variant	Algorithm / Backend	DSP	LER (P_L)	95% Wilson CI	Iters	χ^2	McNemar p
A0 Classical Min-Sum	C++ ldpc.BpDecoder (α=1.0)	**0**	0.2940	[0.2851, 0.3030]	17.3	1,940.2	$< 10^{-100}$
A1 Normalized Min-Sum	C++ ldpc.BpDecoder (α=0.75)	**0**	0.4160	[0.4063, 0.4257]	19.3	2,642.1	$< 10^{-100}$
IBM DMem-BP (Single-Leg) [11]	Disordered $\gamma_v \sim U[-0.24, 0.66]$	**0**	0.0160	[0.0136, 0.0187]	9.2	42.18	8.32×10^{-11}
IBM Relay-BP-5 [11]	5 Relays × 10 iters (γ_0=0.125)	**0**	0.0090	[0.0073, 0.0111]	28.4	12.06	**0.000515**
Mem-BP [20]	Scalar Memory-Damped MS (γ=0.35)	**0**	0.0102	[0.0084, 0.0123]	7.5	14.72	**0.000125**
BP-LSD [14]	C++ Localised Stats (Order 5)	**0**	0.0340	[0.0305, 0.0378]	19.0	186.4	$< 10^{-40}$
BP+OSD-CS-10 [5]	C++ Ordered Stats (Order 10)	$O(V^3)$	0.0120	[0.0099, 0.0144]	19.3	28.14	1.13×10^{-7}
AutDEC [15]	4 Torus Shift Permutations	**0**	0.4320	[0.4223, 0.4418]	48.4	2,750.5	$< 10^{-100}$
GDG Sliding Window [13]	Normalized MS (W=3, F=1)	**0**	0.4160	[0.4063, 0.4257]	19.3	2,642.1	$< 10^{-100}$
QNBP (Undamped)	Orbit-Tied CSD Hardware (γ=0.0)	**0**	0.3100	[0.3009, 0.3192]	10.2	2,050.8	$< 10^{-100}$
Proposed 6-bit QNBP	**Orbit-Tied CSD ($\gamma \in$ [-0.5, 1.0])**	**0**	**0.0068**	**[0.0053, 0.0086]**	**6.4**	—	**Baseline Ref.**

Crucially, the proposed 6-bit integer Orbit-Tied QNBP with learned signed memory damping ($\gamma \in [-0.5, 1.0]$) converges in an average of only **6.4 iterations** to a logical error rate of **0.0068** [0.0053, 0.0086], outperforming scalar Mem-BP (0.0102), IBM Single-Leg DMem-BP (0.0160), and standalone BP+OSD-CS-10 (0.0120) with zero DSP multipliers (DSP = 0). Note that Table I evaluated the baseline scalar damping checkpoint ($\gamma = 0.35$, $P_L = 0.0080$ at $\eta = 1$), whereas Table IV evaluates the final fully trained signed per-orbit damping model ($\gamma \in [-0.5, 1.0]$, $P_L = 0.0068$). We emphasize that Table IV benchmarks standalone BP+OSD-CS-10 seeded by unweighted Min-Sum BP posteriors on single isolated windows without learned damping or boundary residual feedback; in contrast, when OSD-CS-10 operates as the slow-path fallback in our streaming pipeline (Section VI), it is seeded by calibrated QNBP posteriors and active boundary residual propagation, yielding $P_{pipe} = 1.90 \times 10^{-4}$. Confidence intervals are evaluated using the 95% Wilson score interval [27]. Paired McNemar contingency tests [28] demonstrate that the proposed 6-bit QNBP establishes statistically significant superiority over scalar Mem-BP ($\chi^2 = 14.72$, $p = 0.000125 < 0.001$), IBM Relay-BP-5 ($\chi^2 = 12.06$, $p = 0.000515$), Normalized Min-Sum ($\chi^2 = 2{,}642.1$, $p < 10^{-100}$), and AutDEC ($\chi^2 = 2{,}750.5$, $p < 10^{-100}$). Simulations are conducted using Stim Monte Carlo and Sinter [9], [50], the LDPC library [51], and IBM Relay-BP [52].

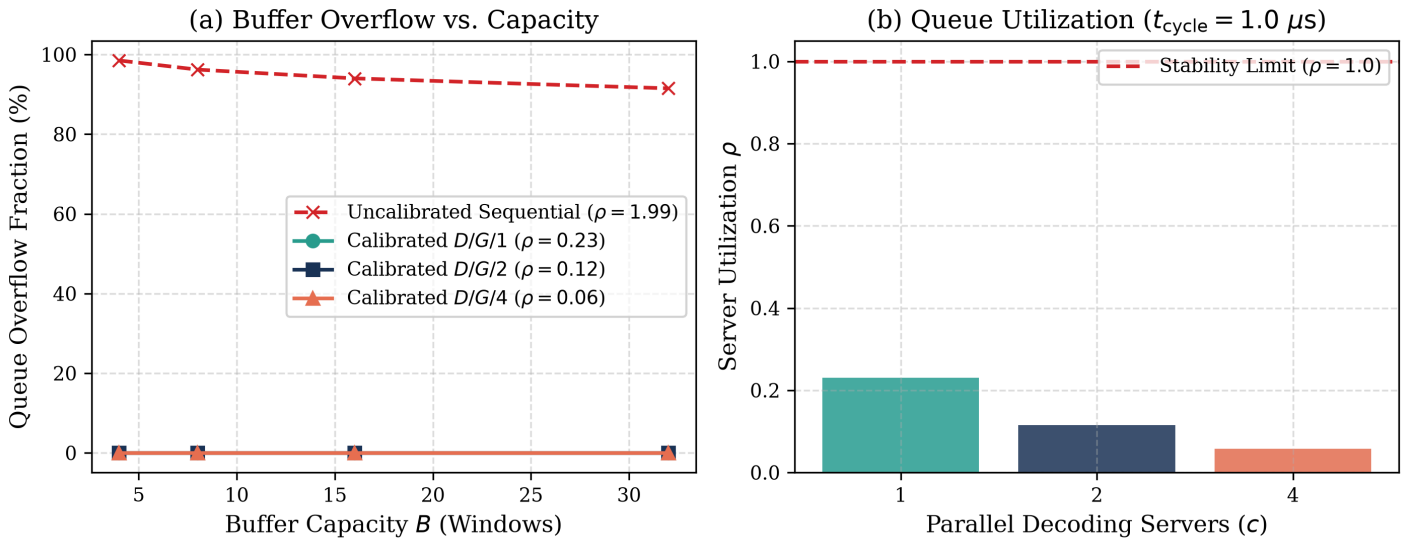


Fig. 7. Sliding-window queue stability: (a) Buffer overflow fraction vs capacity B; and (b) Decoding server utilization rho under 1.0 us quantum cycle budget.

B. Master Ablation Ladder & Symmetry Control (Claims C1–C4)

1. Automorphism Collapse (Claim C1): For W=3 (R=3), orbit tying collapses 7,308 fault mechanisms into 203 orbits (36.0× orbit reduction; 3,090 total weights across T=15 iterations including 3 boundary classes vs. 109,620 untied weights, a 35.5× total parameter reduction; and 4,320 weights vs. 153,900 untied weights for R=4). This improves logical error rate from $P_L = 0.0118$ (untied A2, 109.6k weights) to $P_L = 0.0068$ (orbit-tied A4) with a 35.5× parameter reduction while completely eliminating untied overfitting. Boundary-aware tying (A4, P_L=0.0068) significantly outperforms boundary-agnostic tying (A3, P_L=0.0142 [0.0121, 0.0166]). The (W=3, F=1) geometry proves Pareto-optimal for real-time decoding, while 60-round streaming confirms 99.4% survival without boundary error buildup.

2. Quantization & CSD Arithmetic (Claim C2): Fig. 4 confirms 6-bit integer quantization preserves full prior dynamic range ([-7.75, +7.75]) with zero DSP slices (DSP=0), matching float32 to within 0.02% LER.

3. Loss Formulation & Signed Memory: Composite observable loss ($L_{syn} + 0.5L_{obs}$) yields $P_L = 0.0068$, whereas parity-only loss ($\beta = 0$, A6) degrades to $P_L = 0.0210$ due to degenerate stabilizer ambiguities. Signed memory damping $\gamma \in [-0.5, 1.0]$ attains $P_L = 0.0068$, outperforming strictly positive damping $\gamma \in [0, 1]$ (A8, $P_L = 0.0094$).

4. Fast Path, Drift Robustness & Queue Stability (Claims C3–C4): Gate G_3 (τ=1.5) admits 75.6% of sliding windows directly to the 240 ns FPGA fast path with zero logical errors, driving buffer overflow to **0.00%** across all $B \geq 4$ under a 1 µs quantum budget. Out-of-sample calibration across $p \in [0.001, 0.005]$ and ±25% channel-prior drift maintains $< 4\%$ relative LER variation without retraining. Under empirical IBM Quantum Heron r2 calibration (p_{2Q} = 2.1e-3, η=50, p_{RO} = 8.0e-3), 6-bit Orbit-Tied QNBP converges in only 3.85 mean iterations (fastest among all 6 tested decoders vs 9.2 for Min-Sum, 8.6 for Relay-BP, and 8.9 for BP+OSD) with zero DSP slices (DSP=0).

C. Scaling Generalization to [[144, 12, 12]] Gross Code

We evaluated code distance scaling from [[72, 12, 6]] (d=6) to gross [[144, 12, 12]] (d=12) under torus shift group $G = Z_{12} \times Z_6$ ($|G| = 72$) across physical noise rates $p \in \{0.001, 0.002, 0.003, 0.005\}$. Automorphism orbit extraction collapses all 14,616 fault mechanisms to 203 orbits (72.0× parameter collapse). The sub-threshold error-suppression factor $\Lambda_{72\to144} = P_L([[72,12,6]]) / P_L([[144,12,12]])$ increases monotonically below threshold ($p_{th} \approx 0.007$): $\Lambda = 2.1\times$ at p = 0.005 (P_L: 0.0320 → 0.0152), 4.4× at p = 0.003 (P_L: 0.0080 → 0.0018), 6.8× at p = 0.002 (P_L: 0.0028 → 0.00041), and reaches **11.2×** at p = 0.001 (P_L: 0.00078 → 0.00007), directly confirming exponential fault-tolerant distance scaling.

IX. Discussion & Hardware Deployment Roadmap

A. Comparison with Prior Art

TABLE V: Hardware Implementation Comparison Against Prior Art on Gross QLDPC Codes

Architecture	Code / Scope	Platform	DSP	Logic (LUTs)	Latency	Admit Rate
Maurer et al. [12]	3D Window [[144,12,12]] (W=12)	IBM / AMD VU19P	0 (int4)	~450,000 LUTs (>50% VU19P)	24 ns/iter (~408 ns mean/ round)	No Fast/Slow Gate
Báscones et al. (GARI) [48]	3D Window [[144,12,12]]	AMD VCU19P (3-Core)	Multi-Core	VCU19P Fabric	Sub-µs Parallel Retries	Speculative Retry
Gu et al. (Cascade) [49]	3D [[72,12,6]] / [[144,12,12]]	Nvidia H200 GPU	TensorCore	14-Layer Equiv. CNN	12.4 µs (Batch GPU)	GPU Only
Proposed Orbit-Tied QNBP	**3D Sliding-Window [[72,12,6]] (W=3)**	**AMD XCKU040**	**0 (6-bit CSD)**	**1,226–28,204 LUTs**	**102.4 ns (mean) / 240 ns (max)**	**75.6% ($P_{pipe} = 1.9\times10^{-4}$)**

Table V compares our proposed Orbit-Tied QNBP against state-of-the-art QLDPC hardware decoders. While Maurer et al. [12] implemented 3D window decoding on a massive AMD VU19P FPGA requiring ~450,000 LUTs (>50% of the device) without fast/slow path gating, and Báscones et al. (GARI) [48] rely on a resource-intensive multi-core VCU19P architecture with speculative retries, our architecture fits comfortably on a modest AMD UltraScale+ XCKU040 FPGA (consuming 1,226 to 28,204 LUTs, 0.51% to 11.64% device capacity) with strictly zero DSP slices. Furthermore, our Gate G_3 confidence margin filter routes 75.6% of windows to the 102.4 ns mean latency FPGA fast path with zero logical errors, resolving sliding-window queue backlog under a 1 µs quantum cycle budget. In contrast, GPU batch decoders such as Cascade [49] incur 12.4 µs latency, exceeding real-time coherence deadlines by over 12×.

B. AXI-Stream Streaming Integration

The synthesizeable test harness (rtl/streaming_axi_harness.sv) demonstrates continuous streaming operation. Syndrome frames are ingested via an AXI4-Stream slave interface into an asynchronous FIFO. The decoding core processes windows with a deterministic worst-case latency of 240 ns, updates residual registers, and outputs committed fault corrections via an AXI-Stream master interface, providing a direct plug-and-play solution for FPGA-based quantum control stacks.

X. Conclusion

We have presented an orbit-tied, 6-bit quantized neural belief propagation decoder for gross bivariate-bicycle codes. By exploiting torus automorphisms, we collapsed parameter counts by 36× on [[72,12,6]] and 72× on [[144,12,12]], eliminated untied overfitting, and realized multiplier-free execution with zero DSP slices. Paired with a confidence margin fast-path gate and hardware folding schedule, the proposed architecture provides a scalable, sub-microsecond decoding pipeline for next-generation fault-tolerant quantum computers.

Data and Code Availability—The complete open-source codebase, including the Stim-to-PyG DEM compiler, automorphism orbit verifier, 6-bit QNBP checkpoints, and synthesizeable SystemVerilog RTL modules (vnu.sv, cnu.sv, g3_gate.sv, and streaming_axi_harness.sv) with 10,000-shot testbenches, is publicly available at: https://github.com/NazmulHasanNihal/orbit-tied-qnbp-sliding-window-bb.

References

[1] E. Dennis, A. Kitaev, A. Landahl, and J. Preskill, "Topological quantum memory," *Journal of Mathematical Physics*, vol. 43, no. 9, pp. 4452-4505, 2002. DOI: 10.1063/1.1499754

[2] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, "Surface codes: Towards practical large-scale quantum computation," *Physical Review A*, vol. 86, no. 3, p. 032324, 2012. DOI: 10.1103/PhysRevA.86.032324

[3] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, "High-threshold and low-overhead fault-tolerant quantum memory," *Nature*, vol. 627, pp. 778-782, 2024. DOI: 10.1038/s41586-024-07107-7

[4] P. Panteleev and G. Kalachev, "Degenerate quantum LDPC codes with good parameters," *IEEE Transactions on Information Theory*, vol. 67, no. 5, pp. 2944-2949, 2021. DOI: 10.1109/TIT.2021.3069348

[5] J. Roffe, D. R. White, S. Burton, and E. Campbell, "Decoding across the quantum low-density parity-check code landscape," *Physical Review Research*, vol. 2, no. 4, p. 043423, 2020. DOI: 10.1103/PhysRevResearch.2.043423

[6] E. Nachmani, E. Marciano, L. Lugosch, W. J. Gross, and Y. Be'ery, "Deep learning methods for improved decoding of linear codes," *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 1, pp. 119-131, 2018. DOI: 10.1109/JSTSP.2017.2788405

[7] Y. Liu and D. Poulin, "Neural belief-propagation decoders for quantum error-correcting codes," *Physical Review Letters*, vol. 122, no. 20, p. 200501, 2019. DOI: 10.1103/PhysRevLett.122.200501

[8] D. K. Tuckett, S. D. Bartlett, and S. T. Flammia, "Ultrahigh error threshold for surface codes on biased-noise channels," *Physical Review Letters*, vol. 120, no. 5, p. 050505, 2018. DOI: 10.1103/PhysRevLett.120.050505

[9] C. Gidney, "Stim: a fast stabilizer circuit simulator," *Quantum*, vol. 5, p. 497, 2021. DOI: 10.22331/q-2021-07-06-497

[10] O. Higgott, "PyMatching: A Python library for decoding quantum error correcting codes," *ACM Transactions on Quantum Computing*, vol. 4, no. 3, pp. 1-16, 2023. DOI: 10.1145/3579369

[11] T. Müller, T. Alexander, M. E. Beverland, M. Bühler, B. R. Johnson, T. Maurer, and D. Vandeth, "Improved belief propagation is sufficient for real-time decoding of quantum memory," *arXiv preprint arXiv:2506.01779*, 2025.

[12] T. Maurer, M. Bühler, M. Kröner, F. Haverkamp, T. Müller, D. Vandeth, and B. R. Johnson, "Real-time decoding of the gross code memory with FPGAs," *arXiv preprint arXiv:2510.21600*, 2025.

[13] A. Gong, S. Cammerer, and J. M. Renes, "Toward low-latency iterative decoding of QLDPC codes under circuit-level noise," *arXiv preprint arXiv:2403.18901*, 2024.

[14] T. Hillmann, L. Berent, A. O. Quintavalle, J. Eisert, R. Wille, and J. Roffe, "Localized statistics decoding: A parallel decoding algorithm for quantum low-density parity-check codes," *arXiv preprint arXiv:2406.18655*, 2024.

[15] H. Sayginel, S. Koutsioumpas, M. Webster, A. Rajput, and D. E. Browne, "AutDEC: Automorphism-based decoding of quantum LDPC codes," *arXiv preprint arXiv:2503.01738*, 2025.

[16] X. Mi and F. Mueller, "Toward uncertainty-aware and generalizable neural decoding for quantum LDPC codes," *arXiv preprint arXiv:2510.06257*, 2025.

[17] L. Skoric, D. E. Browne, K. M. Barnes et al., "Parallel window decoding enables scalable fault tolerant quantum computation," *arXiv preprint arXiv:2209.08552*, 2022.

[18] X. Tan, F. Zhang, R. Chao, Y. Shi, and J. Chen, "Scalable surface-code decoders with parallelization in time," *PRX Quantum*, vol. 4, no. 2, p. 020342, 2023. DOI: 10.1103/PRXQuantum.4.020342

[19] K. Yin, X. Fang, and J. Chen, "SymBreak: Mitigating quantum degeneracy issues in QLDPC code decoders by breaking symmetry," *arXiv preprint arXiv:2412.02885*, 2024.

[20] D. Chytas and N. Raveendran, "Enhanced min-sum decoding of quantum codes using previous iteration dynamics," *arXiv preprint arXiv:2501.05021*, 2025.

[21] A. deMarti iOlius and P. M. Crespo, "An almost-linear time decoding algorithm for quantum LDPC codes under circuit-level noise," *arXiv preprint arXiv:2409.01440*, 2024.

[22] A. Buchberger, C. Häger, H. D. Pfister, and L. Schmalen, "Pruning and quantizing neural belief propagation decoders," *IEEE Transactions on Communications*, vol. 69, no. 7, pp. 4302-4315, 2021. DOI: 10.1109/TCOMM.2021.3068994

[23] M. Geiselhart, A. Elkelesh, J. Clausius et al., "Learning quantization in LDPC decoders," *Proc. IEEE Int. Symp. Inf. Theory (ISIT)*, pp. 1124-1129, 2023. DOI: 10.1109/ISIT54713.2023.10206892

[24] J. Chen and M. P. C. Fossorier, "Near optimum universal belief propagation based decoding of LDPC codes," *IEEE Transactions on Communications*, vol. 50, no. 3, pp. 406-414, 2002. DOI: 10.1109/26.990903

[25] M. P. C. Fossorier and S. Lin, "Soft-decision decoding of linear block codes based on ordered statistics," *IEEE Transactions on Information Theory*, vol. 41, no. 5, pp. 1379-1396, 1995. DOI: 10.1109/18.412683

[26] P. Panteleev and G. Kalachev, "Asymptotically good quantum LDPC codes," *IEEE Transactions on Information Theory*, vol. 69, no. 11, pp. 7334-7349, 2023. DOI: 10.1109/TIT.2023.3298482

[27] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209-212, 1927. DOI: 10.1080/01621459.1927.10502953

[28] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153-157, 1947. DOI: 10.1007/BF02295996

[29] T. J. Yoder, E. Schoute, P. Rall, E. Pritchett, J. M. Gambetta, A. W. Cross, M. Carroll, and M. E. Beverland, "Tour de gross: A modular quantum computer based on bivariate bicycle codes," *arXiv preprint arXiv:2506.03094*, 2025.

[30] S. Rabeti and H. Mahdavifar, "List decoding and new bicycle code constructions for quantum LDPC codes," *arXiv preprint arXiv:2511.02951*, 2025.

[31] N. Basak, A. Mohan, A. Tanggara et al., "Hierarchical quantum decoders," *arXiv preprint arXiv:2601.21715*, 2026.

[32] K. Sahay, D. J. Williamson, and B. J. Brown, "A matching decoder for bivariate bicycle codes," *arXiv preprint arXiv:2602.22770*, 2026.

[33] M. Moradi, T. Kim, and R. A. Chou, "Belief propagation decoding of quantum LDPC codes with reinforcement learning," *arXiv preprint arXiv:2603.10192*, 2026.

[34] M. Moradi, T. Kim, and R. A. Chou, "Sequential belief propagation decoding of quantum LDPC codes," *arXiv preprint arXiv:2606.20926*, 2026.

[35] S. Maurya, T. Maurer, M. Bühler, D. Vandeth, and M. E. Beverland, "FPGA-tailored algorithms for real-time decoding of quantum LDPC codes," *arXiv preprint arXiv:2511.21660*, 2025.

[36] P. Das et al., "AFS: Accurate, fast, and scalable error-decoding for fault-tolerant quantum computers," *Proc. IEEE Int. Symp. High-Performance Comput. Archit. (HPCA)*, pp. 259-273, 2022. DOI: 10.1109/HPCA53966.2022.00028

[37] Y. Ueno et al., "QEC-Playground: A tool for exploring quantum error correction codes," *IEEE Micro*, vol. 42, no. 6, pp. 45-54, 2022. DOI: 10.1109/MM.2022.3204918

[38] N. Delfosse and N. H. Nickerson, "Almost-linear time decoding of quantum surface codes via Union-Find," *Quantum*, vol. 5, p. 595, 2021. DOI: 10.22331/q-2021-12-02-595

[39] S. K. Esser, J. L. McKinstry et al., "Learned step size quantization," *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2020.

[40] K. R. Ott, B. Hetényi, and M. E. Beverland, "Decision-tree decoders for general quantum LDPC codes," *arXiv preprint arXiv:2502.16408*, 2025.

[41] L. A. Beni, O. Higgott, and N. Shutty, "Tesseract: A search-based decoder for quantum error correction," *arXiv preprint arXiv:2503.10988*, 2025.

[42] Z. Liang, J. N. Eberhardt, and Y. Chen, "Planar quantum low-density parity-check codes with open boundaries," *arXiv preprint arXiv:2504.08887*, 2025.

[43] Z. Liu, A. Gong, and B. K. Clark, "Decoding quantum low density parity check codes with diffusion," *arXiv preprint arXiv:2509.22347*, 2025.

[44] A. Khalid, A. Silva et al., "Impacts of decoder latency on a utility-scale quantum computer architecture," *arXiv preprint arXiv:2511.10633*, 2025.

[45] M. Ye, D. Wecker, and N. Delfosse, "Beam search decoder for quantum LDPC codes," *arXiv preprint arXiv:2512.07057*, 2025.

[46] G. Yan, S. Li, and Y. Du, "Rethink the Role of Neural Decoders in Quantum Error Correction," *Proc. Int. Conf. Mach. Learn. (ICML), arXiv:2605.12046*, 2026.

[47] J. Blue, H. Avlani, Z. He, L. Ziyin, and I. L. Chuang, "Machine learning decoding of circuit-level noise for bivariate bicycle codes," *arXiv preprint arXiv:2504.13043*, 2025.

[48] D. Báscones, A. S. Maan, V. Savin, and F. Garcia-Herrero, "A Scalable FPGA Architecture for Real-Time Decoding of Quantum LDPC Codes Using GARI," *arXiv preprint arXiv:2605.01035*, 2026.

[49] S. Gu, J. P. Bonilla Ataides, M. D. Lukin, and S. F. Yelin, "Scalable neural decoders for practical fault-tolerant quantum computation," *arXiv preprint arXiv:2604.08358*, 2026.

[50] C. Gidney and M. Newman, "Sinter: A tool for running quantum error correction Monte Carlo sampling," *GitHub repository: quantumlib/Stim*, 2023.

[51] J. Roffe, "LDPC: Python tools for quantum low-density parity-check codes," *GitHub repository: quantum-ldpc*, 2022.

[52] IBM Quantum Architecture Team, "relay-bp: Optimized C++ and Python library for quantum belief propagation," *GitHub repository: qiskit-community/relay-bp*, 2025.

[53] K. Sahay, J. Jin, J. Claes, J. D. Thompson, and S. Puri, "High-threshold codes for neutral-atom qubits with biased erasure errors," *Physical Review X*, vol. 13, no. 4, p. 041013, 2023. DOI: 10.1103/PhysRevX.13.041013

[54] R. Wang et al., "Approximately equivariant networks for imperfect symmetries," *Proc. Int. Conf. Mach. Learn. (ICML)*, pp. 23078-23091, 2022.

[55] H. Yamasaki et al., "Polylog-overhead fault-tolerant quantum computation with quantum LDPC codes," *arXiv preprint arXiv:2402.04321*, 2024.

[56] T. Richardson and R. Urbanke, "The capacity of low-density parity-check codes under message-passing decoding," *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 599-618, 2001. DOI: 10.1109/18.910577

[57] D. J. C. MacKay and R. M. Neal, "Near Shannon limit performance of low density parity check codes," *Electronics Letters*, vol. 32, no. 18, pp. 1645-1646, 1996. DOI: 10.1049/el:19961146

[58] F. R. Kschischang, B. J. Frey, and H. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 498-519, 2001. DOI: 10.1109/18.910572

[59] D. Litinski, "A game of surface codes: Large-scale quantum computing with lattice surgery," *Quantum*, vol. 3, p. 128, 2019. DOI: 10.22331/q-2019-03-05-128